

Irreducibility in $\mathbb{K}[x]$.

- **Exercise 1**

Determine polynomials below is rather reducible or not.

	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{C}$
$p_1(x) = x^2 + 4x + 4 = (x + 2)^2$	1	1	1	1
$p_2(x) = x^2 - 4 = (x - 2)(x + 2)$	1	1	1	1
$p_3(x) = 9x^2 - 3 = 3(\sqrt{3}x - 1)(\sqrt{3}x + 1)$	1	0	1	1
$p_4(x) = x^2 - \frac{4}{9} = \left(x - \frac{2}{3}\right)\left(x + \frac{2}{3}\right)$	0	1	1	1
$p_5(x) = x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2})$	0	0	1	1
$p_6(x) = x^2 + 1 = (x - i)(x + i)$	0	0	0	1

- **Exercise 2**

Prove that $x = 1$ is a root of $p(x) = 2x^4 - 3x^3 - 3x^2 + 7x - 3$ and find its multiplicity. Is $p(x)$ reducible over $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$?

- **Solution 2.1**

$p(1) = 2 - 3 - 3 + 7 - 3 = 0 \implies x = 1$ is a root. Then $p(x) = (x - 1)(2x^3 - x^2 - 4x + 3)$.

We find that $2 - 1 - 4 + 3 = 0$. Then $p(x) = (x - 1)^2(2x^2 + x - 3)$ and then $p(x) = (x - 1)^3(2x + 3)$. Therefore p is reducible over $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

- **Exercise 3**

Find a real root of $f(x) = x^3 - \sqrt{2}x^2 + x - \sqrt{2}$.

- **Solution 3.1**

We have that $f(x) = (x - \sqrt{2})(x^2 + 1)$. Therefore $x = \sqrt{2}$ is a real root.

- **Exercise 4**

Determine $p(x) = x^{12} + x^6 + 1$ is reducible or not over $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

- **Solution 4.1**

We have that

$$\begin{aligned} p(x) &= x^{12} + 2x^6 + 1 - x^6 \\ &= (x^6 + 1)^2 - x^6 \\ &= (x^6 + 1 - x^3)(x^6 + 1 + x^3). \end{aligned}$$

Therefore it is reducible over $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

- **Exercise 5**

Find the roots of the following polynomial $p(x) = x^5 - 3x^4 + 5x^3 - x^2 - 10x$.

- **Solution 5.1**

We have that $p(x) = x(x^4 - 3x^3 + 5x^2 - x - 10)$. By rational roots theorem, possible rational roots are $\pm 1, \pm 2, \pm 5, \pm 10$. We find that $p(-1) = p(2) = 0$. Thus $p(x) = x(x + 1)(x + 2)(x^2 - 2x + 5)$.

- **Exercise 6**

Let $p(x) = x^5 + a_3x^3 + a_2x^2 + a_1x + a_0$ be the polynomial with real coefficients. Suppose that $x_1 = 1 + i$ is one root of multiplicity 2. Prove that $a_0 = 16$.

- **Solution 6.1**

$1 - i$ is also a root of multiplicity 2. Then $(x - (1 + i))^2(x - (1 - i))^2 = x^4 - 4x^3 + 8x^2 - 8x + 4 \mid p(x)$. Suppose that $p(x) = (x + r)(x^4 - 4x^3 + 8x^2 - 8x + 4) = x^5 + (r - 4)x^4 + (8 - 4r)x^3 + (8 - 8r)x^2 + (4 - 8r)x + 4r$. Therefore $a_0 = 4r = 16$.